Harmonic-Deviation Scoring for Spatially Clustered Heritage Data:
A Pipeline and Worked Example with UNESCO World Heritage Sites

Seth Tenenbaum*

Independent Scholar, USA

Abstract

Spatial analyses of heritage data routinely impose coordinate-dependent scoring functions, yet seldom report how the choice of reference frame interacts with the geographic clustering typical of cultural corpora. This paper contributes a reproducible analytical pipeline — from unit discovery through anchor characterization, within-frame diagnostics, invariant boundary tests, and cross-corpus stability checks — for detecting, characterizing, and delimiting harmonic-deviation structure in spatially clustered datasets. The pipeline is demonstrated on the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$). A parameter sweep over candidate spacings and reference longitudes selects a frame near 3° spacing and the ∼35°E corridor. Within-frame diagnostics record consistent concentration at harmonic nodes, including cluster asymmetry and stronger behavior among domed/spherical monuments. A full-corpus Rayleigh test returns null ($p_{\text{perm}} = 0.3716$), establishing that the structure is frame-specific, not invariant periodicity. The pipeline's key empirical output is a cross-corpus stability finding: the ∼35°E / 3° frame produces elevated enrichment not only in UNESCO but also in two independently assembled Eurasian datasets (Wikidata Q180987 stupas, $N = 229$; OWTRAD Silk Road nodes, $N = 1674$) under identical frozen parameters, a consistency that does not appear when the same pipeline is run at other reference longitudes. Whether this cross-corpus stability reflects Eurasian corridor geography, something specific to the selected spacing, or both remains an open empirical question; the priority follow-up is application of the frozen pipeline to heritage corpora outside the Eurasian corridor

*ORCID: 0009-0008-5797-2498. Correspondence: seth@fourthtemple.com. The author declares no competing interests.

by an independent team. All code, data, and reproduction scripts are deposited at Zenodo.

Keywords: spatial statistics, harmonic-deviation scoring, cultural analytics, UNESCO World Heritage, cross-corpus stability, monument distribution, representation dependence, analytical pipeline

1 Introduction

Spatial analysis of heritage data depends on choices that are rarely made explicit: the coordinate system, the reference frame, the scoring or distance function, and the scale at which proximity is evaluated. When a corpus is geographically clustered — as most cultural heritage datasets are — these choices interact with the clustering geometry and can produce stable, statistically detectable within-frame patterns that do not appear under invariant tests. Whether such patterns are artefacts of the scoring procedure, reflections of the underlying geography, or evidence of something more specific is a question that existing spatial methods do not systematically address.

Prior work on spatial periodicity in monument distributions (Thom, 1967; Kendall, 1974; Freeman, 1976) was criticised for investigator-assembled site lists, ad hoc unit choices, and no principled way to separate frame-dependent structure from invariant periodicity. More recent probability-based frameworks for archaeoastronomical alignments (Magli, 2013, 2016; Ruggles, 1999, 2015; Silva & Steele, 2020) and the ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) have advanced the field but have not addressed the specific problem of how to detect, characterise, and delimit harmonic-deviation structure across multiple corpora.

This paper contributes a reproducible analytical pipeline that addresses each of these failure modes. The pipeline uses externally curated corpora, parameter sweeps for unit and anchor, tier thresholds derived from a held-out corpus, explicit boundary-condition tests that distinguish frame-specific structure from invariant periodicity, and a cross-corpus stability check that asks whether the selected frame organises related datasets under frozen parameters.

The pipeline is demonstrated on the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$), with stability checks against the Wikidata Q180987 stupa corpus ($N = 229$) and the OWTRAD attested Silk Road network ($N = 1674$ unique endpoints). The worked example produces a specific empirical output: a frame near 3° spacing and the $\sim$35°E corridor that shows cross-corpus stability across all three datasets — a consistency that does not generalise to other reference longitudes. The pipeline correctly identifies that this structure is frame-specific (the Rayleigh test is null) and that the cross-corpus stability, while genuine, may reflect shared Eurasian corridor geography rather than an invariant spatial period. Determining which explanation is correct requires application of the frozen pipeline to non-Eurasian heritage corpora, a step we identify as the priority follow-up.

Paper structure. Section 2 describes the pipeline in general terms. Section 3 describes the corpora. Section 4 presents the worked example, organised by pipeline stage. Section 5 reports the cross-corpus stability finding that is the pipeline's key empirical output. Section 6 interprets what the pipeline reveals. Section 7 draws implications for cultural analytics practice. Section 8 concludes.

2 The Pipeline

The pipeline consists of seven stages, described here in general terms and instantiated for the worked example in Sections 4–5.

Stage 1: Unit discovery. A sweep over candidate angular spacings ω identifies which spacing produces the strongest node-proximity enrichment in the primary corpus. A degree-denominated Monte Carlo null tests whether the selected spacing exceeds what geographic clustering alone would produce.

Stage 2: Anchor characterisation. For the selected ω, a global sweep over candidate reference longitudes ϕ (at fine resolution across $[0°, 360°\text{E})$) identifies the corridor that maximises within-frame enrichment. A permutation null calibrates whether the selected corridor is globally unusual.

Stage 3: Tier-threshold derivation. Proximity tiers are defined by angular thresholds, ideally derived from a held-out corpus's angular dispersion so that the tier widths do not depend on the primary data. This stage fixes the vocabulary (A++, A+, A, B, C) used by the diagnostics.

Stage 4: Within-frame diagnostics. Conditional on the selected (ω, ϕ) frame, a battery of tests characterises the within-frame structure. Three threshold-free diagnostics form the primary family (Holm step-down adjusted): a von Mises mixture test on a morphologically defined sub-corpus, a Spearman cluster-asymmetry test, and a phase-rotation null. Supporting enrichment tests provide secondary summaries. All reported p-values at this stage are conditional on the selected frame.

Stage 5: Boundary-condition tests. At least one invariant test — one that does not assume the selected reference frame — checks whether the within-frame structure extends to invariant periodicity. Disagreement (as in the worked example) locates the result as frame-specific structure, not invariant periodicity.

Stage 6: Geographic-concentration nulls. Targeted null models test whether the within-frame enrichment in sub-populations exceeds what their longitude footprint alone would produce. A within-subgroup bootstrap (Null A) and a restricted geographic draw (Null C) address this question from complementary directions.

Stage 7: Cross-corpus stability. The selected frame is applied with all parameters frozen to independently assembled corpora in the same hypothesised domain. The same diagnostics are run. Agreement supports promotion of the candidate frame to "worth investigating further" status; disagreement would demote it. This stage also includes a comparison to alternative reference longitudes, which tests whether the cross-corpus stability is specific to the selected corridor or would appear at many anchors.

3 Data

3.1 Primary Corpus: UNESCO World Heritage

The UNESCO World Heritage XML dataset (UNESCO, 2024) contains 1248 inscribed nominations. Retaining Cultural and Mixed sites (excluding 235 Natural-only) and dropping 2 entries lacking geocoordinates yields $N = 1011$ sites. The corpus over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe and North America: 50%; Asia-Pacific: 23%; Latin America and Caribbean: 11%; Arab States: 9%; Africa: 7%). This geographic concentration is itself a structural feature that interacts with harmonic scoring; the pipeline is designed to characterise rather than assume away that interaction.

Some UNESCO properties are serial or multi-component nominations; the XML dataset provides a single representative coordinate. The scoring therefore partly reflects the coordinate structure of the corpus as published.

3.2 Stability-Check Corpora

Two independently assembled datasets serve as cross-corpus stability checks (Stage 7).

Wikidata Q180987 stupas ($N = 229$). A corpus of stupa sites retrieved from Wikidata (Tenenbaum, 2026). It overlaps geographically and culturally with UNESCO's South and Southeast Asian inscriptions but was assembled for a different purpose and by a different process.

OWTRAD Silk Road network ($N = 1674$ unique endpoints). The Old World Trade Routes database (Ciolek, 2004), comprising 1946 route segments. This corpus represents historical trade connectivity rather than monumental architecture.

3.3 Domed-Monument Sub-corpus

A keyword filter (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) applied to UNESCO site names and description texts yields $N = 90$ sites. The keyword set was refined iteratively during analysis; results from this sub-corpus are hypothesis-generating. A context-validated subset ($N = 83$, 7 ambiguous sites rejected; Appendix C) is used for sensitivity checks.

4 Worked Example: UNESCO Heritage Corpus

4.1 Stage 1: Unit Discovery

A sweep over candidate spacings from 0.5° to 15° identifies a spacing near 3° as the highest-scoring parameter. Adjacent non-harmonic spacings (1.8°, 2.1°, 2.7°, 3.3°, 3.6°) do not reach significance. The 1.5° and 6° spacings also score well, but both are mathematical consequences of the 3° structure (every 3° node is a 1.5° node; 6° nodes are a strict subset). A Monte Carlo unit sweep is consistent with 3° as the dominant period.

The 3° value is treated throughout as the pipeline's selected analytic spacing, not as a demonstrated cultural, historical, or metrological unit.

4.2 Stage 2: Anchor Characterisation

For the selected spacing near 3°, a global sweep over 36,000 trial anchors (0° to 360°E, 0.01° resolution) identifies the ∼35°E longitude corridor. Both Mount Gerizim (35.269°E; 132 A+ sites, 99.7$^{\text{th}}$ percentile) and Jerusalem (35.232°E; 128 A+, 97.7$^{\text{th}}$ percentile) place the corridor in the global top 2.3% of trial anchors. The relevant empirical object is the ∼35°E corridor, not either landmark individually; we label by Gerizim because it is not inscribed in the UNESCO list and therefore does not self-exclude from the corpus.

Full anchor-sweep permutation calibration yields $p = 0.0753$ (†). This marginal result is a boundary condition: the selected frame is used as an exploratory representation for conditional diagnostics, not as a globally significant discovery from the full parameter search.

Candidate anchors outside the corridor do not show comparable enrichment. Mecca ($p = 0.2825$, ns), Mt. Meru ($p = 0.3956$, ns), and Mt. Kailash ($p = 0.3956$, ns) are all non-significant. This anchor specificity becomes important at Stage 7.

4.3 Stage 3: Tier Thresholds

Sites are assigned to descriptive tiers by their deviation δ from the nearest harmonic node:

$$\delta_i = \left|(\lambda_i - \phi) \bmod \omega\right|, \quad \text{folded to } [0, \omega/2].$$

Tier-A++ ($\delta \leq 0.06°$, $\lesssim 6.66$ km; null rate 4%); Tier-A+ ($\delta \leq 0.15°$, $\lesssim 16.6$ km; null rate 10%); Tier-A ($\delta \leq 0.3°$, $\lesssim 33$ km; null rate 20%). Kilometer labels are equatorial

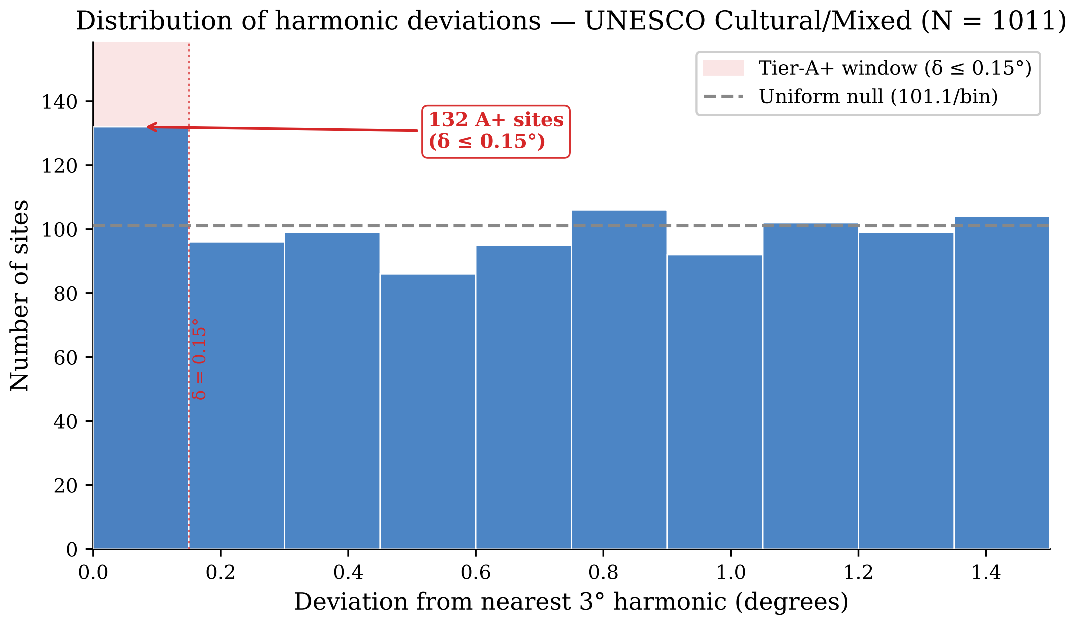

Figure 1: Harmonic deviation distribution for the full UNESCO corpus ($N = 1011$) under the selected 3° frame. The shaded region marks the Tier-A+ window ($\delta \leq 0.15°$); the dashed line marks the 10% geometric null. The pipeline's Stage 5 (Section 4.5) tests whether this excess persists under invariant analysis.

approximations; no latitude correction is applied. In this worked example, tier widths are derived from the von Mises dispersion of the Wikidata stupa corpus ($\hat{\sigma} = 0.1922°$, $\hat{\kappa} = 6.17$), so they do not depend on the UNESCO data (Appendix B).

4.4 Stage 4: Within-Frame Diagnostics

All results in this subsection are conditional on the selected (ω, ϕ) frame. The three primary diagnostics are Holm step-down adjusted ($k = 3$).

Empirical observation. The full corpus has a Tier-A+ rate of 13.1% (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), against the 10% geometric null (Figure 1). The enrichment is 1.31× the null rate.

Test 1 (dome von Mises mixture). A two-component von Mises/uniform mixture fit to the dome sub-corpus ($N = 90$) yields $\hat{w} = 0.0876$ ($\approx 8.8\%$ of dome sites concentrated at the selected phase; $\hat{\kappa} = 252.19$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). Because the dome keyword set was iteratively refined, this test is exploratory subset characterisation.

Test 2 (Spearman cluster asymmetry). $\rho = -0.1076$, permutation $p = 0.00065$ (***; Holm $p_{\text{adj}} = 0.002$, **). Nodes drawing more sites show smaller mean deviations, the expected signature when the scoring function places nodes near cluster centres.

Test 3 (phase-rotation null). The selected reference phase minimises the full-corpus mean harmonic deviation relative to random grid placements ($p_{\text{perm}} = 0.04729$, *; Holm $p_{\text{adj}} = 0.047$, *).

Table 1: Stage 4 diagnostics: within-frame tests conditional on the selected 3° / ∼35°E frame.

Test	N	Raw p	Holm p_{adj}	Role
Test 1 (dome VM mix.)	90	0.00180[b]	0.004 **	Exploratory subset
Test 2 (Spearman ρ)	1011	0.00065[b]	0.002 **	Full-corpus diagnostic
Test 3 (mean dev.)	1011	0.04729[b]	0.047 *	Full-corpus diagnostic

[b]permutation or bootstrap p. Holm (1979) step-down (Holm, 1979).

Dome sub-population detail. As an illustrative subset result, 18/90 dome sites fall at Tier-A+ (20.0%, 2.00×; Fisher $p = 0.0346$ *). Tier-A++ concentration is stronger: 11/90 (12.2%, 3.06× the 4% null; permutation $p = 0.004$ **; bootstrap $p = 0.006$ **).

Leave-n-out sensitivity. No single A+ site drives the dome result: removing any A+ site yields $N = 89$, A+$= 17$, $p = 0.0067$. Leave-3-out worst case across all $\binom{18}{3}$ triples: $p = 0.0253$. The A++ sub-population: leave-3-out worst case $p = 0.0233$ (leave-2-out: $p = 0.0087$).

4.5 Stage 5: Boundary-Condition Tests

The full-corpus Rayleigh test for 3° periodicity returns $R = 0.0224$ ($Z = 1.06$, $p_{\mathrm{perm}} = 0.3716$ ns): no spectral periodicity at 3° is detectable in the raw longitude distribution. This fixes the boundary: the within-frame structure identified at Stage 4 is frame-specific, not invariant periodicity.

A pipeline that produced uniformly positive results at every stage would be suspect. The Rayleigh null is evidence that the pipeline correctly reports the limits of the candidate frame: the within-frame diagnostics are positive, the invariant test is negative, and the pipeline presents both without collapsing the distinction.

A ±0.3% shift from the selected 3° spacing collapses joint within-frame concentration across the dome sub-population, confirming that the within-frame structure is sharply phase-dependent.

4.6 Stage 6: Geographic-Concentration Nulls

Domed monuments are concentrated in the Eurasian corridor (71% in 20° to 120°E versus 41% of the full corpus). Two null models ask whether this footprint explains the within-frame enrichment.

Null A (within-dome bootstrap). Drawing $N = 90$ longitudes from the dome sites' own longitude list yields a mean expected A+ count consistent with the observed value ($p = 0.5429$, ns). The dome enrichment is consistent with where dome sites already are.

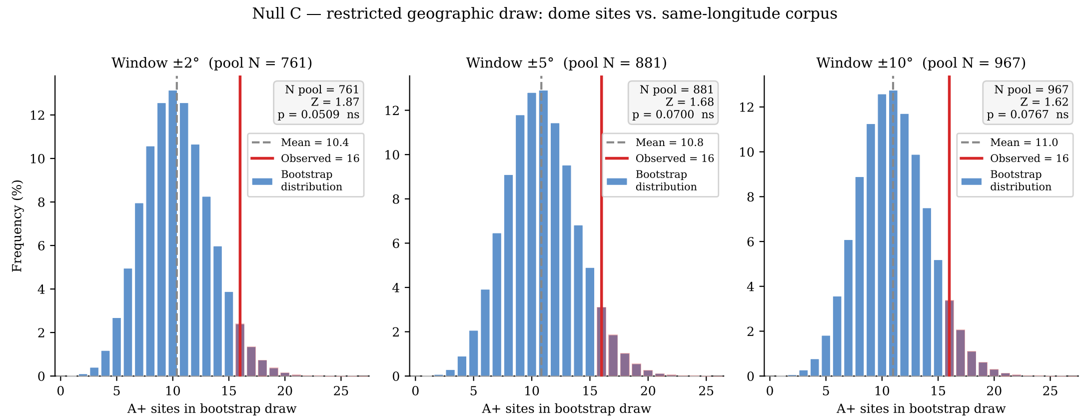

Figure 2: Null C bootstrap distributions for the context-validated dome sub-population ($N = 83$, observed A+ $= 16$) at three window widths.

Null C (restricted geographic draw). Drawing from all full-corpus sites within $\pm W^\circ$ of any dome longitude, the dome subset nominally exceeds the broader footprint at all three tested windows: $\pm 2^\circ$ ($p = 0.0257$, *); $\pm 5^\circ$ ($p = 0.0393$, *); $\pm 10^\circ$ ($p = 0.0442$, *). These marginal results do not survive correction across the three windows.

The pipeline reports both results without forcing a resolution: Null A is consistent with geographic concentration being sufficient; Null C suggests possible residual morphological specificity but is marginal. This is the kind of ambiguity the pipeline is designed to surface rather than suppress.

Americas negative control. The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, below the 10% null ($p = 0.5149$ ns, one-sided depletion). The within-frame pattern is absent outside the Eurasian corridor.

5 Cross-Corpus Stability: The Pipeline's Key Output

Stage 7 is the pipeline's most informative diagnostic. It asks: does the selected frame, with all parameters frozen, produce comparable enrichment in independently assembled datasets? And is that stability specific to the selected corridor, or would it appear at many reference longitudes?

5.1 Stability Under Frozen Parameters

Wikidata Q180987 stupas. Under the identical anchor, spacing, and tier thresholds, $N = 229$ stupas yield 70/229 Tier-A sites (30.6%, 1.53× null; binomial $p < 0.001$ ***; permutation $p_{\text{perm}} = 0.065$ $\sim$). Harmonic nodes with at least one Tier-A stupa carry 2.56× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry

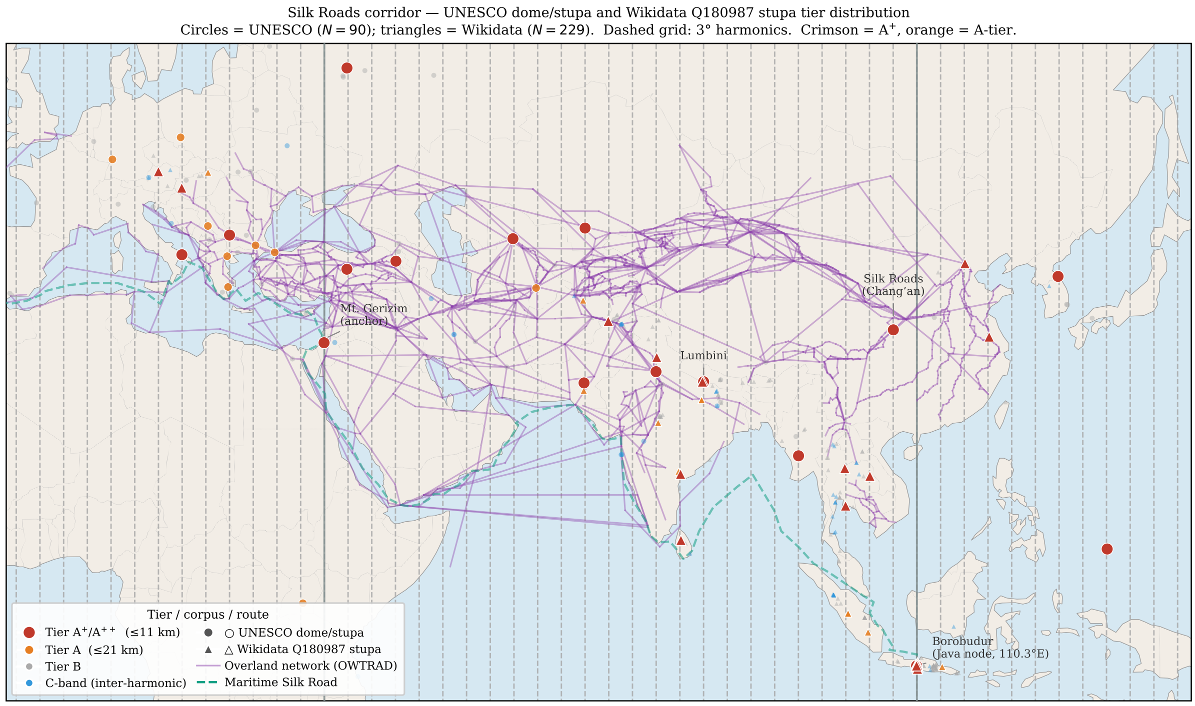

Figure 3: Tier distribution of UNESCO dome/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor, overlaid on OWTRAD attested trade routes (Ciolek, 2004). The 3° harmonic grid (dashed) is the selected frame from Stage 2; the figure shows where the frozen parameters produce consistent cross-corpus organisation.

result of Test 2. The Java/Sumatra node (105° to 115°E) concentrates 42/54 sites at Tier-A (77.8%; OR = 18.38×, $p < 0.001$ ***).

OWTRAD Silk Road network. Testing all 3892 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 211 fall at A++ harmonics, 457 at A+, and 886 at A (1.36× null; $z = 4.53$, $p < 0.0001$ *** for A++; $p = 0.0002$ *** for A+). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0202$ *, consistent with enrichment specific to the selected phase. Harmonics containing at least one A++ endpoint (30 harmonics) carry 2.70× more route connectivity than harmonics without one (11 harmonics; $p = 0.0007$ ***, Mann-Whitney one-tailed).

5.2 Corridor Specificity

The cross-corpus stability is specific to the ∼35°E corridor. Both Gerizim (132 A+, 99.7$^{\text{th}}$ percentile) and Jerusalem (128 A+, 97.7$^{\text{th}}$ percentile) place the corridor in the global top 2.3% of 36,000 trial anchors. Candidate anchors outside the corridor — Mecca (107 A+, $p = 0.2825$, ns), Mt. Meru (104 A+, $p = 0.3956$, ns), Mt. Kailash (104 A+, $p = 0.3956$, ns) — do not produce comparable enrichment in the UNESCO corpus, let alone the same cross-corpus stability in the Wikidata and OWTRAD data.

This is the answer to the natural objection that any reference longitude would produce comparable structure. It would not. The ∼35°E corridor is distinguished not by the fact that harmonic-deviation scoring found enrichment in a single corpus — it would do so at many anchors — but by the cross-corpus stability of that enrichment across three independently assembled datasets under identical parameters.

5.3 Scope of the Stability Finding

All three corpora are concentrated in the Eurasian corridor. The cross-corpus stability is therefore consistent with two explanations: (1) the selected frame captures something specific to the 3° spacing and ∼35°E phase, or (2) the three corpora share enough geographic structure that any frame fitting one will fit the others. The pipeline cannot distinguish these with the available data. The Americas negative control (Section 4.6) suggests the pattern does not extend outside the Eurasian corridor, but a sample of $N = 145$ is too small to be definitive.

What *would* distinguish the two explanations is application of the frozen pipeline to heritage corpora assembled outside the Eurasian corridor — for example, national registers from sub-Saharan Africa, the Pacific, or Mesoamerica — by an independent team using the deposited code and parameters. If the 3° / ∼35°E frame shows elevated enrichment in non-Eurasian data, the first explanation gains support; if not, the stability reflects shared corridor geography.

6 Discussion

6.1 What the Pipeline Reveals

The worked example separates three things that prior studies of spatial periodicity have tended to conflate.

First, within-frame concentration: the selected frame organises the UNESCO corpus and two related datasets in a way that random permutations of the same frame do not (Tests 1–3, all significant after Holm correction). This is a genuine property of the data under the chosen representation.

Second, the absence of invariant periodicity: the full-corpus Rayleigh test is null ($p_{\text{perm}} = 0.3716$). The within-frame concentration does not extend to an invariant 3° period in raw longitude. The pipeline distinguishes these two statuses by design.

Third, cross-corpus stability that is corridor-specific: the same frame produces elevated enrichment in three independently assembled Eurasian datasets but not at other reference longitudes. This is the empirical observation that makes the selected frame interesting rather than arbitrary.

6.2 The Mechanism

The within-frame concentration arises from the interaction of geographic clustering, the harmonic-deviation scoring function, and frame selection. The UNESCO corpus is clustered along the Eurasian corridor; the parameter sweep selects a frame that places lattice nodes near dense regions; the scoring function rewards proximity to those nodes. This interaction is sufficient to produce stable within-frame structure without any assumption about intentional placement, hidden geometry, or cultural metrology.

What the mechanism does not fully explain is why the ∼35°E corridor and 3° spacing show elevated enrichment *simultaneously* in three corpora assembled for different purposes. Geographic channelling of Eurasian cultural activity along a common corridor is a plausible partial explanation; whether it is sufficient is an empirical question the present analysis cannot resolve.

6.3 Comparison to Prior Approaches

The pipeline addresses specific weaknesses of prior work. Thom (1967) and Kendall (1974) used investigator-assembled site lists; this pipeline uses externally curated corpora. Freeman (1976) criticised ad hoc unit choices; this pipeline sweeps over candidate units and reports the selection. Prior studies rarely distinguished frame-dependent structure from invariant periodicity; the pipeline's Stage 5 makes that distinction explicit. None of the prior approaches included a cross-corpus stability check at frozen parameters.

7 Implications for Practice

The worked example yields four practical recommendations for spatial analysis of cultural heritage data.

Report the search. State how many frames, phases, spacings, anchors, or thresholds were considered. Distinguish frame selection from subsequent diagnostics. When the same data select the frame and test it, the resulting p-values are conditional on the selected frame.

Include invariant comparisons. A coordinate-aligned result should be checked against at least one method that does not assume the same reference frame. Disagreement locates the result as frame-specific structure.

Test cross-corpus stability at fixed parameters. When multiple corpora are available, freeze the selected frame and apply it to each. Report which corpora show consistent enrichment and which do not. Compare to alternative reference points to assess corridor specificity.

Audit corpus structure. Geographic clustering, corpus overlap, and iterative keyword selection should be treated as part of the inferential design. When multiple corpora share geographic structure, cross-corpus consistency is shared-representation stability, not independent replication.

8 Conclusion

This paper contributes a reproducible analytical pipeline for detecting, characterising, and delimiting harmonic-deviation structure in spatially clustered heritage data. The pipeline's seven stages — unit discovery, anchor characterisation, tier-threshold derivation, within-frame diagnostics, boundary-condition tests, geographic-concentration nulls, and cross-corpus stability checks — are demonstrated on the UNESCO Cultural and Mixed corpus with stability checks against Wikidata Q180987 stupas and the OWTRAD Silk Road network.

The pipeline's key empirical output is a cross-corpus stability finding: a frame near 3° spacing and the ∼35°E corridor produces elevated enrichment across all three independently assembled Eurasian datasets under frozen parameters, a consistency that does not appear at other reference longitudes. The pipeline also correctly identifies what the finding is *not*: the full-corpus Rayleigh test returns null ($p_{\text{perm}} = 0.3716$), and the structure is frame-specific, not invariant 3° periodicity.

Whether the cross-corpus stability reflects Eurasian corridor geography under a particular representational choice, something specific to the 3° spacing, or both remains open. The priority follow-up is application of the frozen pipeline — with all parameters fixed at the deposited values — to heritage corpora outside the Eurasian corridor by an independent team. All code, data, and reproduction instructions are archived at Zenodo (DOI: https://doi.org/10.5281/zenodo.19938283, v1.6.0).

Data and code availability. All analysis scripts, data files, and reproduction instructions are archived at Zenodo (DOI: https://doi.org/10.5281/zenodo.19938283, v1.6.0). The pipeline regenerates all statistics and LaTeX macros from raw data without manual editing (`bash manuscript/reproduce_all_macros.sh`). Full audit files, keyword lists, and supplementary figures are provided in Supplementary S1.

A Tier-Threshold Sensitivity

A log-scale null-rate sweep across 73 thresholds shows that the dome within-frame pattern is not limited to the specific tier width: 57 of 73 thresholds reach $p < 0.05$ for the dome subset; dome OR ranges 1.53×–3.40× across the three canonical tiers. Within ±1 log-decade of the A+ threshold, 27 of 28 thresholds meet the nominal $p < 0.05$ threshold.

B Von Mises Mixture Characterisation

A von Mises + uniform mixture fitted to the harmonic phase deviations of the stability-check corpora under the selected frame provides within-frame dispersion estimates used for tier-threshold derivation (Stage 3): Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922°$. OWTRAD route nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = 0.1543°$. N-weighted pooled $\hat{\sigma} = 0.1588°$; κ ratio $= 1.6$ across corpora. Tier boundaries correspond to $0.25\hat{\sigma}$, $0.50\hat{\sigma}$, and $1.00\hat{\sigma}$ of the stupa corpus dispersion.

C Keyword Audit

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) matched as whole words against site names, XML descriptions, and extended OUV text. Context-validation for ambiguous terms: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Seven sites rejected in the context-validated subset: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru–Kata Tjuṭa, and Diquís Stone Spheres. Full site-by-site audit in the Zenodo archive (Tenenbaum, 2026).

References

Aveni, A. F. (2001). *Skywatchers: A Revised and Updated Version of Skywatchers of Ancient Mexico*. University of Texas Press.

Aveni, A. F. (2003). Archaeoastronomy in the ancient Americas. *Journal of Archaeological Research*, 11(2), 149–191.

Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.

Casson, L. (1989). *The Periplus Maris Erythraei*. Princeton University Press.

Ciolek, T. M. (2004). *Old World Trade Routes (OWTRAD) Project*. Asia Pacific Research Online. http://www.ciolek.com/OWTRAD/DATA/tmcOWTRAD.html

Crown, A. D. (1989). *The Samaritans*. J. C. B. Mohr.

Freeman, P. R. (1976). A Bayesian analysis of the Megalithic yard. *Journal of the Royal Statistical Society, Series A*, 139(1), 20–55.

Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

Hunger, H. & Steele, J. (Eds.) (2019). *The Babylonian World*. Routledge.

Kendall, D. G. (1974). Hunting quanta. *Philosophical Transactions of the Royal Society A*, 276(1257), 231–266.

MacDonald, J. (1964). *The Theology of the Samaritans*. SCM Press.

Magli, G. (2013). *Architecture, Astronomy and Sacred Landscape in Ancient Egypt*. Cambridge University Press.

Magli, G. (2016). *Archaeoastronomy: Introduction to the Science of Stars and Stones*. Springer.

Miksic, J. (1990). *Borobudur: Golden Tales of the Buddhas*. Shambhala.

Powell, M. A. (1987). Masse und Gewichte. In D. O. Edzard (Ed.), *Reallexikon der Assyriologie*, 7, 457–516.

Purvis, J. D. (1968). *The Samaritan Pentateuch and the Origin of the Samaritan Sect*. Harvard University Press.

Ruggles, C. L. N. (1999). *Astronomy in Prehistoric Britain and Ireland*. Yale University Press.

Ruggles, C. L. N. & Cotte, M. (Eds.) (2010). *Heritage Sites of Astronomy and Archaeoastronomy in the Context of the UNESCO World Heritage Convention: A Thematic Study*. ICOMOS / IAU.

Ruggles, C. L. N. (Ed.) (2015). *Handbook of Archaeoastronomy and Ethnoastronomy*. Springer.

Sen, T. (2014). The spread of Buddhism to China: A re-examination of the Buddhist interactions between Ancient India and China. *China Report*, 48(1–2), 11–27.

Silva, F. & Steele, J. (2020). Statistical approaches to detecting archaeoastronomical alignments. In C. Ruggles (Ed.), *Handbook of Archaeoastronomy and Ethnoastronomy*, 295–306. Springer.

Tenenbaum, S. (2026). Harmonic alignment audit data: UNESCO World Heritage, Wikidata stupas, and OWTRAD route network [Data set]. Zenodo. https://doi.org/10.5281/zenodo.19938283

Thom, A. (1967). *Megalithic Sites in Britain*. Oxford University Press.

Tomber, R. (2008). *Indo-Roman Trade: From Pots to Pepper*. Duckworth.

UNESCO World Heritage Centre (2024). World Heritage List [XML dataset]. https://whc.unesco.org/en/syndication